

Chapter 1

BJT Basics I

Device Physics, Active Mode & Small-Signal Modeling

*Based on Razavi, Fundamentals of Microelectronics, 2nd Ed. (Chapters 4–5)
With insights from Razavi's BJT YouTube Lectures (13–28)*

Topics Covered:

- NPN Structure & Terminal Conventions
- Forward Active Mode Operation
- Collector Current Derivation from First Principles
- Base & Emitter Currents, Current Gain β
- Concept of Transconductance (g_m)
- Small-Signal Model (g_m , r_π)
- Early Effect & Output Resistance r_O

Chapter 1: Quick Reference Formula Sheet

Collector Current

Name	Formula	Notes
I_C	$I_C = I_S \cdot \exp(V_{BE}/V_T)$	Active mode, $V_{BC} < 0$
I_S	$I_S = \frac{A_E q D_n n_i^2}{N_B W_B}$	Saturation current
V_T	$V_T = kT/q \approx 26 \text{ mV at } 300\text{K}$	Thermal voltage

Terminal Currents

Name	Formula	Notes
I_B	$I_B = I_C/\beta$	β = current gain (50–200)
I_E	$I_E = I_C + I_B = I_C \cdot \left(1 + \frac{1}{\beta}\right)$	KCL at transistor
α	$\alpha = \frac{\beta}{\beta+1} \approx 1$	For large β , $I_C \approx I_E$

Transconductance

Name	Formula	Notes
g_m	$g_m = I_C/V_T$	Slope of I_C vs V_{BE}
g_m (shortcut)	$g_m = \frac{1}{26\Omega}$ at $I_C = 1\text{mA}$	Very useful form for exams

Small-Signal Model

Name	Formula	Notes
r_π	$r_\pi = \beta/g_m = V_T/I_B$	Base-emitter resistance
$g_m v_\pi$	$\Delta I_C = g_m \cdot v_\pi$	Controlled source

Early Effect

Name	Formula	Notes
I_C (Early)	$I_C = I_S \exp(V_{BE}/V_T) \left(1 + \frac{V_{CE}}{V_A}\right)$	V_A = Early voltage
r_O	$r_O = V_A/I_C$	Output resistance
$A_{v,\max}$	$A_v = -g_m \cdot (R_C \parallel r_O)$	Gain is limited by r_O

1 The Bipolar Transistor: What It Is and Why It Matters

At its core, a bipolar junction transistor (BJT) is a **voltage-controlled current source**. The voltage across two of its terminals (base-emitter) controls the current flowing between the other two (collector-emitter). This is the single most important idea in this chapter.

Razavi begins Chapter 4 by asking: if we have a device whose output current is proportional to an input voltage, $I_1 = KV_1$, and we pass that current through a load resistor R_L , we get $V_{out} = -KR_L \cdot V_{in}$. If $KR_L > 1$, we have **amplification**. The BJT is precisely such a device, except its current depends *exponentially* on V_{BE} rather than linearly.

LECTURE INSIGHT (Lecture 13)

Razavi emphasizes in Lecture 13 that students should think of the BJT first as a “voltage-to-current converter”—the base-emitter voltage is the input, and the collector current is the output. Everything else (base current, emitter current) is secondary. He says: “The BJT is fundamentally a transconductor. That is its job.”

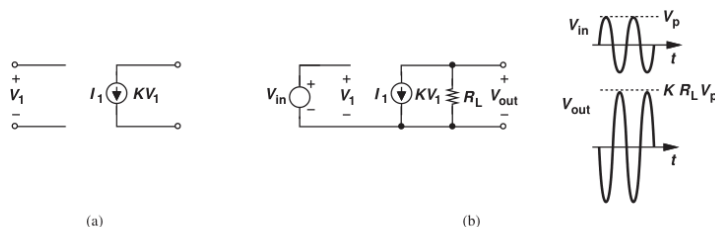


Figure 4.1 (a) Voltage-dependent current source, (b) simple amplifier.

Figure 1: Figure 4.1 from Razavi (a) Voltage-dependent current source, (b) simple amplifier using it.

2 Structure of the NPN Bipolar Transistor

The BJT consists of three doped semiconductor regions forming a sandwich: an **npn** transistor has a p-type base layer between two n-type regions (emitter and collector). The three terminals are the **Base (B)**, **Collector (C)**, and **Emitter (E)**.

Physical picture: The emitter “emits” charge carriers (electrons in npn), the collector “collects” them, and the base “controls” how many make the journey. The emitter is heavily doped (n^+) so that the electron current from $E \rightarrow B$ dominates over the hole current from $B \rightarrow E$. The base must be very thin ($\sim 100 \text{ \AA}$ in modern devices) so most electrons reach the collector without recombining.

Although the device looks symmetric, the emitter and collector are NOT interchangeable—they have different doping levels and dimensions.

EXAM TIP

On exams, always define your voltage polarities clearly. V_{BE} means $V_B - V_E$. A common error is confusing V_{BE} with V_{EB} . For npn in active mode: $V_{BE} > 0$ ($\approx 0.7\text{--}0.8 \text{ V}$) and $V_{BC} < 0$ (collector is more positive than base).

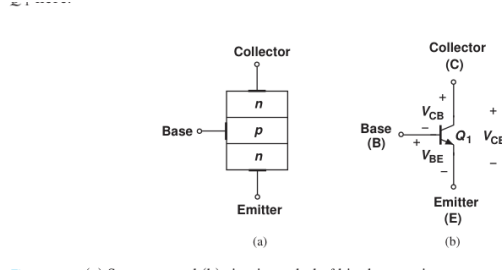


Figure 2: Figure 4.4 from Razavi (a) NPN structure showing three doped regions, (b) circuit symbol.

3 Operation in the Forward Active Mode

Conditions for active mode: Base-emitter junction is **forward-biased** ($V_{BE} > 0$) and base-collector junction is **reverse-biased** ($V_{BC} < 0$, equivalently $V_{CE} > V_{BE}$).

Physical process (step by step):

1. **Forward-biased BE junction:** Because $V_{BE} > 0$ (typically $\sim 0.8V$), electrons are injected from the n^+ emitter into the p-type base.
2. **Diffusion through thin base:** The injected electrons find themselves in the base with a high concentration near the emitter side and zero concentration near the collector side. This concentration gradient drives diffusion of electrons across the base.
3. **Collection by reverse-biased CB junction:** The strong electric field in the CB depletion region sweeps electrons arriving from the base into the collector.
4. **Result:** A large current flows from collector to emitter (conventional direction), controlled by V_{BE} . A small base current (holes) is needed to sustain the process.

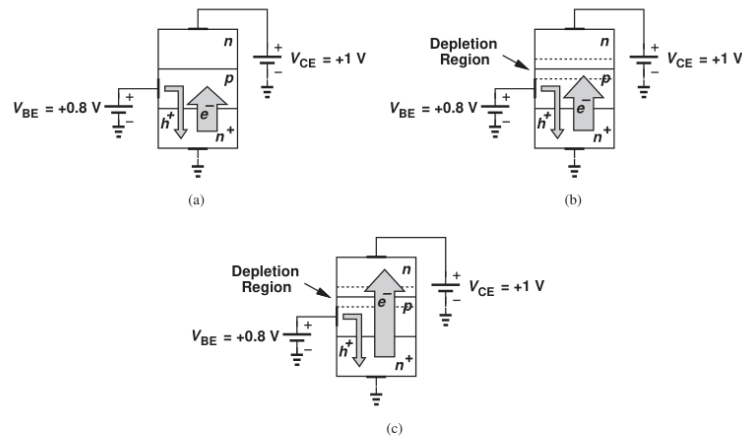


Figure 4.7 (a) Flow of electrons and holes through base-emitter junction, (b) electrons approaching collector junction. (c) electrons passing through collector junction.

Figure 3: Figure 4.7 from Razavi Electron flow from emitter through base to collector.

LECTURE INSIGHT (Lecture 14)

In Lecture 14, Razavi uses the analogy: “Think of the base as a thin hotel lobby. Electrons enter from the emitter door, and most rush straight through to the collector door because the lobby is so narrow. Only a few stop to recombine with holes (the hotel staff). The thinner the lobby, the more electrons make it through.”

4 Derivation of Collector Current

We now derive the most important equation in BJT physics: $I_C = I_S \exp(V_{BE}/V_T)$. Every step has a clear physical origin.

DERIVATION: Collector Current I_C

- 1. Electron concentration at BE junction edge (x_1):** excess minority carrier concentration is:

$$\Delta n(x_1) = \frac{n_i^2}{N_B} \left[\exp\left(\frac{V_{BE}}{V_T}\right) - 1 \right] \quad (\text{Eq. 4.5})$$

- 2. Electron concentration at CB junction edge (x_2):** CB junction is reverse-biased, so any electron reaching x_2 is swept away. Therefore $n(x_2) = 0$.

- 3. Linear concentration profile:** With n high at x_1 and zero at x_2 , the electron density varies approximately linearly across the base width W_B .

- 4. Diffusion current:** Apply Fick's law: $J_n = qD_n \frac{dn}{dx} = qD_n \frac{0 - \Delta n(x_1)}{W_B}$.

- 5. Total current:** Multiply by emitter area A_E and flip sign for conventional current:

$$I_C = \frac{A_E q D_n n_i^2}{N_B W_B} \left[\exp\left(\frac{V_{BE}}{V_T}\right) - 1 \right]$$

- 6. Simplification:** For $V_{BE} \gg V_T$ (which is always true since $\approx 0.8\text{V} \gg 26\text{mV}$), we drop the -1 :

$$I_C = I_S \exp\left(\frac{V_{BE}}{V_T}\right) \quad \text{where} \quad I_S = \frac{A_E q D_n n_i^2}{N_B W_B}$$

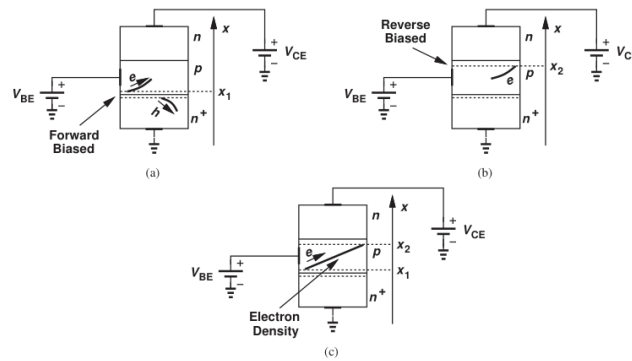


Figure 4.8 (a) Hole and electron profiles at base-emitter junction, (b) zero electron density near collector, (c) electron profile in base.

Figure 4: Figure 4.8 from Razavi Electron density profile in the base.

KEY RESULT:

$$I_C = I_S \exp\left(\frac{V_{BE}}{V_T}\right) \quad \text{where} \quad I_S = \frac{A_E q D_n n_i^2}{N_B W_B}$$

Assumptions: Active mode ($V_{BE} > 0, V_{BC} < 0$); base width W_B constant.

When assumptions break: If $V_{BC} > 0$, saturation mode starts. If V_{CE} changes, the Early effect slightly modifies I_C .

EXAM TIP

Quick calculation: a 60 mV increase in V_{BE} multiplies I_C by $e^{60/26} \approx 10$. So every ~ 60 mV change gives one decade of current change. This “60 mV per decade” rule is very useful on exams.

5 Base Current, Emitter Current, and Current Gain β

Where does the base current come from? Two mechanisms:

1. **Reverse injection:** Holes flow from the base into the emitter. Since $N_E \gg N_B$, this is small.
2. **Recombination:** Some electrons injected into the base recombine with holes before reaching the collector. The base terminal must supply replacement holes.

Both components are small fractions of I_C . We define the **current gain**: $I_C = \beta I_B$, where β typically ranges from 50 to 200.

DERIVATION: Terminal Current Relationships

Start with KCL: $I_E = I_C + I_B$

Since $I_B = I_C / \beta$: $I_E = I_C \left(1 + \frac{1}{\beta}\right) = I_C \frac{\beta+1}{\beta}$

Define $\alpha = \frac{\beta}{\beta+1}$. Then $I_C = \alpha I_E$. For $\beta = 100$, $\alpha = 0.99$, so $I_C \approx I_E$.

KEY RESULT:

$$I_C = \beta I_B \quad | \quad I_E = (\beta + 1) I_B \quad | \quad I_C \approx I_E \quad | \quad \alpha = \frac{\beta}{\beta + 1}$$

Assumptions: Active mode; β constant.

When assumptions break: β varies with temperature and current level. Circuit design should not depend critically on β .

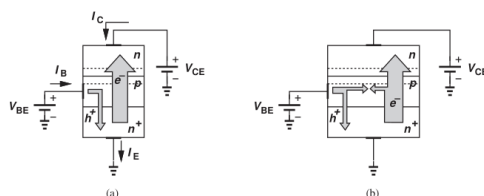


Figure 4.12 Base current resulting from holes (a) crossing to emitter and (b) recombining with electrons.

Figure 5: Figure 4.12 from Razavi Base current from reverse injection and recombination.

LECTURE INSIGHT (Lecture 15)

Razavi stresses in Lecture 15: “Never design a circuit that depends on the exact value of β . β is unreliable it varies from device to device. The collector current I_C is the reliable quantity because it depends on V_{BE} and I_S .”

6 Concept of Transconductance g_m

Physical question: How “strong” is the BJT as a voltage-to-current converter? If we change V_{BE} by a small amount ΔV_{BE} , how much does I_C change?

The answer is the **transconductance**, defined as $g_m = \frac{dI_C}{dV_{BE}}$. It measures the “gain” of the transistor as a voltage-controlled current source.

DERIVATION: Transconductance g_m

$$g_m = \frac{dI_C}{dV_{BE}} = \frac{d}{dV_{BE}} \left[I_S \exp \left(\frac{V_{BE}}{V_T} \right) \right]$$

$$g_m = \frac{1}{V_T} I_S \exp \left(\frac{V_{BE}}{V_T} \right) = \frac{I_C}{V_T}$$

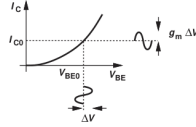


Figure 4.18 Illustration of transconductance.

Figure 6: Figure 4.18 from Razavi Graphical interpretation of g_m as slope.

KEY RESULT:

$$g_m = \frac{I_C}{V_T} \quad (V_T \approx 26 \text{ mV at } 300\text{K})$$

Assumptions: Active mode; small-signal (perturbation $\ll V_T$).

EXAM TIP

Memorize: $g_m = I_C/V_T$. At $I_C = 1\text{mA} \rightarrow g_m = 1/26\Omega$. Think of $1/g_m$ as a resistance of 26Ω at 1mA . For any other current, just scale: at $2\text{mA} \rightarrow 1/g_m = 13\Omega$.

7 The Small-Signal Model

The small-signal model replaces the nonlinear transistor with linear circuit elements, valid for small perturbations around a DC operating point.

DERIVATION: Small-Signal Model Parameters

1. Collector current change: $\Delta I_C = g_m \Delta V_{BE} = g_m v_\pi \rightarrow$ modeled as a controlled source $g_m v_\pi$ from C to E.

2. Base current change: $\Delta I_B = \Delta I_C / \beta = (g_m / \beta) v_\pi \rightarrow$ Since v_π and ΔI_B are between the same two terminals (B and E), this is equivalent to a resistor:

$$r_\pi = \frac{v_\pi}{\Delta I_B} = \frac{\beta}{g_m} = \frac{V_T}{I_B}$$

3. Collector voltage change: For constant V_{BE} , changing V_{CE} does NOT change I_C (ideally). No element between C and E needed yet.

Result: Just two elements: r_π between B and E, and $g_m v_\pi$ source from C to E.

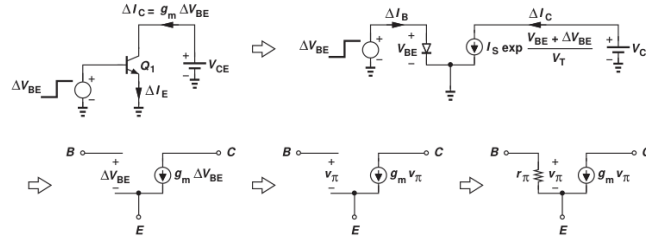


Figure 4.22 Development of small-signal model.

Figure 7: Figure 4.22 from Razavi Development of the BJT small-signal model.

KEY RESULT:

Small-signal model: $r_\pi = \beta / g_m$ between B–E | $g_m v_\pi$ current source from C to E

LECTURE INSIGHT (Lecture 16)

Razavi explains in Lecture 16: “The small-signal model is just a linearization. You find the DC operating point first (bias), then replace the transistor with its small-signal model to analyze the signal behavior. The two analyses—DC and small-signal—are completely separate. Never mix bias quantities with small-signal quantities in the same equation.”

8 The Early Effect and Output Resistance r_O

So far, we said I_C is independent of V_{CE} . In reality, as V_{CE} increases, the collector-base depletion region widens, effectively reducing the active base width W_B . Since $I_C \propto 1/W_B$, a narrower base means higher I_C . This is the **Early effect**.

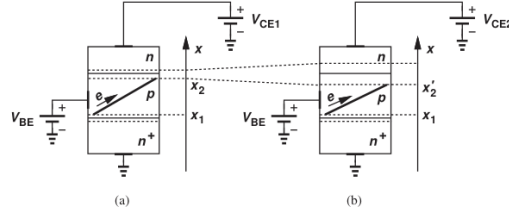


Figure 4.27 (a) Bipolar device with base and collector bias voltages, (b) effect of higher collector voltage.

Figure 8: Figure 4.27 from Razavi Base width modulation.

DERIVATION: Early Effect and r_O

Modified equation: $I_C = I_S \exp(V_{BE}/V_T) \left(1 + \frac{V_{CE}}{V_A}\right)$ where $V_A = \text{Early voltage}$.

Output resistance: $r_O = \left(\frac{\partial I_C}{\partial V_{CE}}\right)^{-1} \approx \frac{V_A}{I_C}$.

This appears in parallel with the $g_m v_\pi$ source in the small-signal model.

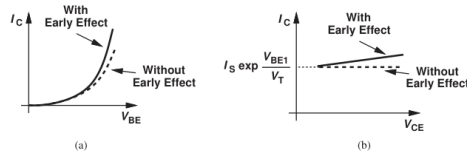


Figure 4.28 Collector current as a function of (a) V_{BE} and (b) V_{CE} with and without Early effect.

Figure 9: Figure 4.28 from Razavi I_C vs V_{CE} characteristics showing lines converging at $-V_A$.

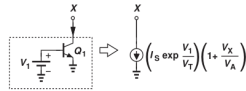


Figure 4.29 Realistic model of bipolar transistor as a current source.

Figure 10: Figure 4.29 from Razavi Small-signal model including r_O .

9 Worked Examples from Razavi

Example 4.5: Computing Output Voltage

Problem: Determine V_{out} in the circuit if $I_S = 5 \times 10^{-16}$ A. Q_1 has $V_{BE} = 750$ mV, $R_L = 1$ k Ω , $V_{CC} = 3$ V.

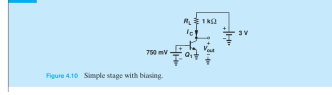


Figure 4.10 from Razavi

Solution:

1. Compute I_C : $I_C = I_S \exp(V_{BE}/V_T) = (5 \times 10^{-16}) \exp(750/26) = \mathbf{1.69}$ mA
2. Voltage drop across R_L : $V_{RL} = I_C R_L = 1.69 \text{ mA} \times 1 \text{ k}\Omega = \mathbf{1.69}$ V
3. Output voltage: $V_{out} = V_{CC} - V_{RL} = 3 - 1.69 = \mathbf{1.31}$ V
4. Verify active mode: $V_C = 1.31$ V, $V_B = 0.75$ V. Since $V_C > V_B$, active mode stands.

Example 4.7: Terminal Currents & Active Mode

Problem: $I_S = 5 \times 10^{-17}$ A, $V_{BE} = 800$ mV, $\beta = 100$, $V_{CC} = 2$ V, $R_C = 500\Omega$. Find all terminal currents and verify active mode. Find max R_C to remain in active mode.

Solution:

- $I_C = 5 \times 10^{-17} \exp(800/26) = \mathbf{1.153}$ mA
- $I_B = I_C/100 = \mathbf{11.53}$ μ A, $I_E = I_B + I_C = \mathbf{1.165}$ mA
- $V_{out} = V_{CC} - I_C R_C = 2 - (1.153 \text{ mA})(500\Omega) = \mathbf{1.424}$ V
- Since $1.424 \text{ V} > 0.8 \text{ V}$, it's in active mode.
- Max R_C : Limit is $V_C = V_B = 0.8 \text{ V}$. So $V_{CC} - I_C R_C = 0.8 \implies R_C = \frac{2-0.8}{1.153\text{m}} = \mathbf{1041\Omega}$.

Example 4.11: Small-Signal Amplification

Problem: I_C is biased to 6.92 mA for a device with $R_C = 100\Omega$. A microphone gives $v_{in} = 1$ mV. Find ΔV_{out} .

Solution:

- $g_m = I_C/V_T = 6.92 \text{ mA}/26 \text{ mV} = \mathbf{266}$ mA/V
- $\Delta I_C = g_m v_{in} = (266 \text{ mA/V})(1 \text{ mV}) = \mathbf{0.267}$ mA
- $\Delta V_{out} = -\Delta I_C R_C = -(0.267 \text{ mA})(100\Omega) = \mathbf{-26.7}$ mV. (Gain ≈ -26.7)

10 Practice Problems

Problem 1. (Based on 4.3)

A BJT has $I_S = 2 \times 10^{-16}$ A and $\beta = 150$. Biased at $V_{BE} = 780$ mV. Find I_C, I_B, I_E, g_m , and r_π .

Solution:

$$I_C = 2 \times 10^{-16} \exp(780/26) = \mathbf{2.138 \text{ mA}}$$

$$I_B = 2.138 \text{ mA}/150 = \mathbf{14.25 \mu\text{A}}$$

$$I_E = 2.138 + 0.01425 = \mathbf{2.152 \text{ mA}}$$

$$g_m = 2.138 \text{ mA}/26 \text{ mV} = \mathbf{82.2 \text{ mA/V}}$$

$$r_\pi = \beta/g_m = 150/82.2 \text{ mS} = \mathbf{1.825 \text{ k}\Omega}$$

Problem 2. (Early Effect)

A BJT has $V_A = 40$ V and is biased at $I_C = 2$ mA. (a) Find r_O . (b) For $R_C = 5 \text{ k}\Omega$, find voltage gain.

Solution:

$$(a) r_O = V_A/I_C = 40\text{V}/2\text{mA} = \mathbf{20 \text{ k}\Omega}$$

$$(b) g_m = 2\text{mA}/26\text{mV} = 76.9 \text{ mA/V}.$$

$$\text{Gain } A_v = -g_m(R_C \parallel r_O) = -76.9 \times (5\text{k} \parallel 20\text{k}) = -76.9 \times 4\text{k} = \mathbf{-307.7}$$

11 Common Mistakes and How to Avoid Them

COMMON MISTAKES

- **X Mistake:** Forgetting to check active mode before using $I_C = I_S \exp(V_{BE}/V_T)$.
OK Fix: Always verify $V_C > V_B$ (for npn) AFTER computing I_C . If $V_C < V_B$, the transistor is in saturation and the regular active region current equations fail!
- **X Mistake:** Using $V_{BE} = 0.7\text{V}$ blindly even when I_S and exact voltages are given.
OK Fix: $V_{BE} = 0.7\text{V}$ is a rule of thumb. For exact calculations yielding gains, always use the exponential equation $I_C = I_S \exp(V_{BE}/V_T)$.
- **X Mistake:** Confusing g_m with $1/r_\pi$.
OK Fix: $g_m = I_C/V_T$ is the transconductance relating v_π to ΔI_C . But $r_\pi = \beta/g_m$ is the base-emitter resistance relating v_π to ΔI_B . They differ by a factor of β .
- **X Mistake:** Mixing DC (bias) and small-signal (AC) quantities.
OK Fix: DC uses uppercase variables (I_C, V_{BE}). Small-signal uses lowercase (i_c, v_{be} or v_π). NEVER add them into the same circuit equation. Bias first, draw small-signal model next.
- **X Mistake:** Forgetting r_O when doing gain analysis with large load resistors.
OK Fix: If R_C is huge, it will be limited by the transistor's own r_O . The maximum intrinsic gain possible is $g_m r_O = V_A/V_T$.

Chapter 1: End-of-Chapter Summary

Physical Constants & Parameters

- **Thermal voltage:** $V_T = kT/q \approx 26 \text{ mV}$ at room temp (300K).
- **Saturation current:** $I_S = \frac{A_E q D_n n_i^2}{N_B W_B}$. Depends heavily on temperature and base width.
- **Current gain β :** Typically 50 to 200. Very unreliable parameter, don't depend on it for biasing!
- **Forward Alpha α :** $\alpha = \frac{\beta}{\beta+1}$. Generally ≈ 0.99 .

DC Active Mode Equations

- **Condition:** $V_{BE} > 0$ (forward-biased BE junction) AND $V_{BC} < 0$ (reverse-biased CB junction).
- **Collector Current:** $I_C = I_S e^{V_{BE}/V_T}$
- **Base & Emitter Current:** $I_B = I_C/\beta$, $I_E = I_B + I_C = I_C \left(\frac{\beta+1}{\beta} \right)$
- **Early Effect:** $I_C(V_{CE}) = I_S e^{V_{BE}/V_T} \left(1 + \frac{V_{CE}}{V_A} \right)$

Small-Signal Parameters (Low Frequency)

To be used **only** after solving the DC operating point to find I_C .

- **Transconductance:** $g_m = \frac{I_C}{V_T}$. Remember: $1/g_m = 26\Omega$ when $I_C = 1 \text{ mA}$.
- **Input Resistance:** $r_\pi = \frac{\beta}{g_m} = \frac{V_T}{I_B}$. Resistance seen looking into the base.
- **Output Resistance:** $r_O = \frac{V_A}{I_C}$. Due to Early effect (base width modulation).
- **Max Intrinsic Gain:** $|A_v|_{\max} = g_m r_O = \frac{V_A}{V_T}$. This is an upper limit on voltage gain.